Guojun Xu

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Research Profile

Ph.D. student in Computer Science and Technology at Wuhan University of Technology, advised by Prof. Junwei Zhou. My research focuses on learned image compression, particularly distributed coding with decoder-side multimodal side information and generative priors for perceptually faithful reconstruction at extremely low bitrates.

Education

Wuhan University of Technology Sep. 2024 – Present

Direct-entry Ph.D. in Computer Science and Technology
Advisor: Junwei Zhou

Wuhan University of Technology Sep. 2020 – Jun. 2024

B.S. in Information and Computing Science

Publications

[1] **Guojun Xu**, Mingyang Zhang, Jianwen Xiang, Cheng Tan, Yanchao Yang, and Junwei Zhou. "Distributed Image Compression with Multimodal Side Information at Extremely Low Bitrates." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. **Accepted.**

[2] **Guojun Xu**, Cheng Tan, Jianwen Xiang, and Junwei Zhou. "CODIC: Distributed Image Compression with Distortion–Perception Co-Optimization." *IEEE Transactions on Multimedia*, 2026. doi:10.1109/TMM.2026.3718254.

Manuscripts Under Review

[3] **Guojun Xu** et al. "Perceptual Distributed Image Compression via Controlled Stochastic Reconstruction." *Data Compression Conference (DCC)*, 2027. **Under Review.**

[4] **Guojun Xu** et al. "Conditional Residual Decomposition for Distributed Image Compression." *International Conference on Learning Representations (ICLR)*, 2027. **Under Review.**

[5] **Guojun Xu** et al. "Retrieval-Augmented Distributed Image Compression with Multi-Reference Side Information." *IEEE Transactions on Image Processing (TIP)*. **Under Review.**

[6] **Guojun Xu** et al. "Distributed Image Compression with Target-Consistent Multimodal Side Information at Extremely Low Bitrates." *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*. **Under Review.**

Research Contributions

Perception–distortion trade-offs [2, 3]

Derived a local Gaussian analysis linking reconstruction scale and source–reconstruction correlation to the distortion cost of stochastic refinement (SPDIC). Developed distortion-oriented and diffusion-based branches with adaptive regional fusion (CODIC).

Multimodal conditioning and side-information reliability [1, 6]

Developed a diffusion decoder combining text derived from correlated images with visual side features. A multimodal alignment objective supervises feature masks for semantic and fine-detail recovery. Extended this direction with reliability-aware conditioning based on bidirectional latent matching and state-aware gating.

Conditional residuals and reconstruction uncertainty [4]

Applied conditional-expectation projection to separate residuals into side-information-predictable and orthogonal components, guiding deterministic correction and stochastic uncertainty modeling.

Retrieval-based multi-reference compression [5]

Retrieved references from a decoder-side gallery using the transmitted representation, then combined depth-aware geometric alignment with multi-reference cross-attention.

Prospective Research Directions

Reliable conditioning for generative compression with imperfect or retrieved side information; uncertainty-aware reconstruction that separates recoverable content from generated variation; evaluation of perceptual quality alongside target-image fidelity.

Awards and Scholarships

Selected Participant, CAST Special Talent Cultivation Project for PhD Students 2026

Cultivating Society: China Society of Image and Graphics (CSIG)

First-class Academic Scholarship for Ph.D. Students 2026

Wuhan University of Technology

Second-class Academic Scholarship for Ph.D. Students 2025

Wuhan University of Technology

First-class Academic Scholarship for Ph.D. Students 2024

Wuhan University of Technology